AYE
TECH HUB

• REVIT BIM • LESSON 03

Project Setup

Templates · Levels · Grids · Units · Project Info

Lesson 03 of 30 | Revit BIM Program

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- Choosing the right Revit template — the decision that shapes everything
- Setting project units, location, and shared coordinates correctly
- Creating levels: the horizontal planes that organise every floor
- Drawing structural grids: the reference lines for all column placement
- Project Information, Title Block setup and issue management

Project Setup — The Foundation of Every Revit Model

The quality of a Revit project is determined in the first ten minutes. A properly set up project — correct template, units, levels, and grids — produces accurate documentation automatically and coordinates seamlessly with other disciplines. A poorly set up project causes errors that compound throughout the entire project lifecycle.

The 6-Step Project Setup Workflow

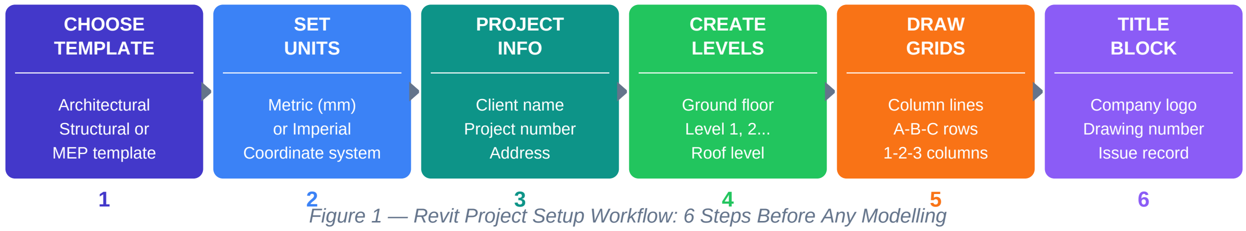

Figure 1 — Revit Project Setup Workflow: 6 Steps Before Any Modelling

Step 1 — Choosing the Right Template

A Revit template (.rte file) is a pre-configured starting point that includes pre-set families, view templates, line styles, annotation types, and project settings. **Never start a project from scratch** — always start from the appropriate template.

Template Type	Use When	Pre-loaded Content
Architectural Template (DefaultMetric.rte)	Designing building form, floor plans, elevations, sections for architecture	Architectural walls, doors, windows, room tags, floor/ceiling/roof families
Structural Template (Structural Analysis.rte)	Designing beams, columns, slabs, foundations, structural framing	Structural framing, column families, analytical model elements, foundation types
MEP Template (Mechanical.rte / Electrical.rte)	Designing HVAC, electrical systems, plumbing — each discipline separate	Duct types, pipe systems, electrical equipment, cable tray, fixtures
Company Template (Custom .rte)	Any project at your firm — contains firm standards and title blocks	Firm logo, standard view templates, project parameters, approved families

Setting	Where to Set	Recommended Value
Length units	Manage → Project Units → Length	Millimetres (mm) for metric
Area units	Manage → Project Units → Area	Square metres (m^2)
Angle units	Manage → Project Units → Angle	Degrees
Project North vs True North	Manage → Position → Rotate True North	Set True North to match site survey
Shared coordinates	Manage → Coordinates → Specify Coordinates	Required for multi-file coordination and site

Project Information: Before modelling anything, fill in Manage → Project Information. This populates the title block automatically throughout the project: project name, number, client, author, address, and issue date. Doing this once here means you never type it manually on a sheet.

Creating Levels — The Skeleton of Your Building

Levels are horizontal reference planes that represent each floor of the building. **Every floor plan, every ceiling plan, and every structural slab is associated with a level**. Getting levels right at the start is critical — changing them later causes widespread disruption to all associated views and elements.

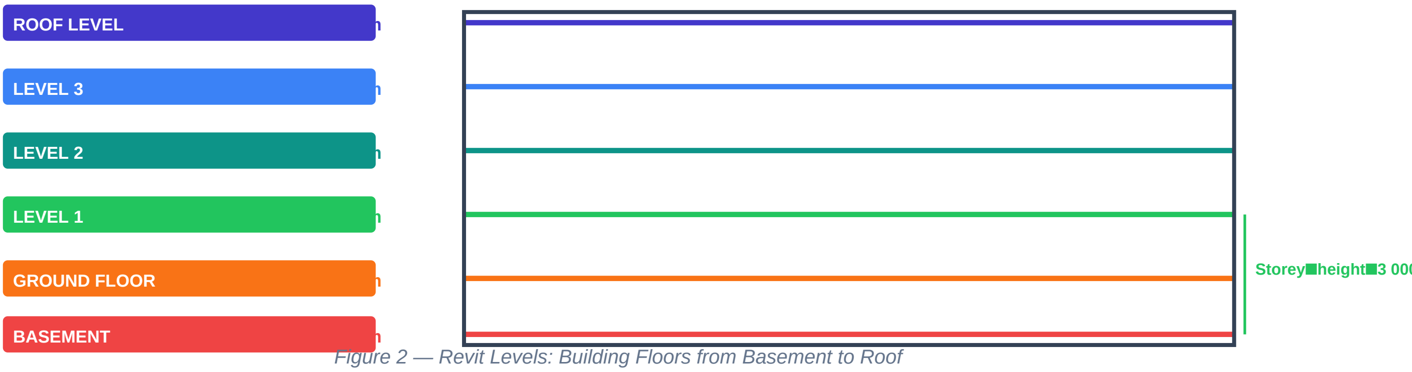

Figure 2 — Revit Levels: Building Floors from Basement to Roof

Level Task	How to Do It in Revit	Key Rule
Create a new level	Open an Elevation view → Architecture tab → Datum panel → Level → Click to place	Levels can ONLY be created in Elevation or Section views, never in plan
Set level elevation	Click the level head number → type new elevation in mm (e.g. 3000 for 3m)	Elevation is measured from the Project Base Point (0,0,0)
Name a level	Click the level name label → type new name (e.g. "Ground Floor", "Level 1")	Name must be unique. Avoid renaming after views and elements are attached.
Copy a level	Select level → Copy (CO) → check "Constrain" for vertical copy → set offset	Copying asks if you want to create an associated plan view — usually YES
Create floor plan from level	View tab → Create → Floor Plan → select the level	If "Create Floor Plan" was not checked on copy, create plan view manually
Check level associations	Select a wall or slab → Properties panel → check "Base Constraint" and "Top Constraint"	All walls and vertical elements must be constrained to levels, not fixed heights

Storey height standard guide: Residential: 2 700–3 000 mm floor-to-floor. Commercial offices: 3 500–4 200 mm. Industrial: 5 000–10 000+ mm. Retail ground floor: 4 500–6 000 mm. Car park: 2 500 mm clearance minimum (2 800 mm floor-to-floor). Always confirm storey heights with the architect before setting levels.

Structural Grids — The Reference Framework

Grids are reference lines that define the structural column and wall grid of a building. **Every column, every structural wall, and every beam should be positioned relative to a grid line.** Grids appear in all plan and elevation views and are shared between Architecture, Structure, and MEP models in a coordinated BIM project.

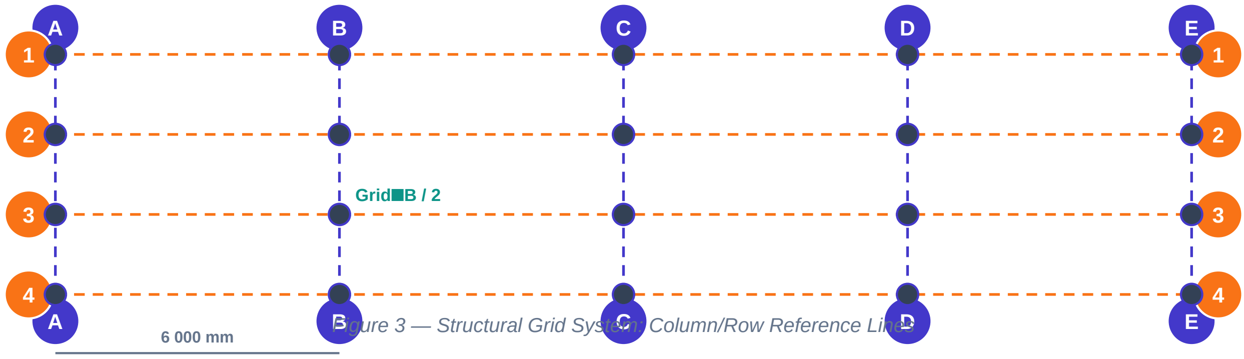

Figure 3 — Structural Grid System: Column/Row Reference Lines

Grid Task	Revit Method	Professional Standard
Draw grid lines	Architecture or Structure tab → Datum → Grid → draw horizontally and vertically	Horizontal grids: numbers (1, 2, 3...) — Vertical grids: letters (A, B, C...)
Set grid spacing	After drawing first grid, use Copy (CO) with specific offset value for uniform bays	Standard bays: 6 000, 7 200, 8 100 mm for commercial; 3 600–5 400 for residential
Rename grid	Click grid head → type new name	Rename to match structural engineer's drawings — critical for coordination
Make grids 3D	Select grid → check "3D" in properties — grid appears in all views and models	3D grids are essential for multi-discipline coordination
Propagate extents	Select grid → right-click → Propagate Extents → select target views	Ensures grids appear at the same length in all plan views
Lock grids	Select grid → Pin (PP shortcut) to prevent accidental movement	Always pin grids and levels once set — they are the project skeleton

Title Blocks and Sheet Setup

Sheet Component	Where to Set	Professional Practice
Title block family	View tab → Sheet Composition → Sheet → select title block	Use company-standard title block (.rfa) loaded into project
Sheet number	Sheet Properties → Sheet Number (e.g. A-001, S-001, M-001)	A=Architectural, S=Structural, M=Mechanical, E=Electrical
Sheet name	Sheet Properties → Sheet Name (e.g. "Ground Floor Plan")	Consistent naming system across all disciplines
Issue date	Manage → Project Information → Issue Date	Updates all title blocks simultaneously — never type on individual sheets
Revision cloud	Annotate → Detail → Revision Cloud — draw around changed area	Add description in Manage → Sheets/Schedule → Revisions → Add row
Drawing register	Create a Sheet List schedule: View → Schedules → Sheet List	Auto-populates with all sheets; updates when sheets are added/renamed

What comes next — REVIT-004: Levels and Grids in Practice — a hands-on session building the complete level and grid framework for a 5-storey commercial building from scratch, including importing a site survey DWG and aligning the Revit model to it.